PERFORMACE FACTORS, ENERGY SYSTEMS AND THEIR APPLICATION

WELCOME

Hello and welcome to this British Athletics - Leadership in Running Fitness Course – performance factors, energy systems and their application.

OBJECTIVES

By the end of this module, you should be able to:

- Describe the factors that influence performance
- Identify the components of fitness
- Identify how training works and differences in energy systems
- Identify how to progress a groups running fitness
- Describe how to adapt a session for a mixed ability group to vary intensity or reflect individual needs

Let's first of all take a look at this video about running a mass participation event.

Take a moment to ask yourself:

- What factors do you think influenced the outcome of the event for the individual runners?
- What factors do you think influenced the runners' performance?

Write down your thoughts – Did you consider any of the following:

How fit the runners are
Their competition nature
How well they slept
What gave them energy (Food & Hydration)
Their running technique

Keep the above thoughts in mind, when we take a look at the 5 fit factors in detail.

INTRODUCTION TO THE 5 PERFORMANCE FACTORS

Here are our 5 performance factors.

Movement ***skills***

Lifestyle and support

Physical ***preparation***

Technical and ***tactical***

Psycho-behavioural

Let's take a look at the performance factors in a bit more detail. Move the athlete along the slider to learn more:

Movement Skills:

Aim to improve:

- Agility
- Balance (static & dynamic)
- Coordination
- Squats, lunges, push, pull, rotate and hinge

Physical Preparation:

Aim to improve or use:

- Speed
- Endurance
- Flexibility
- Acceleration & deceleration
- Reaction time
- Strength & conditioning
- Push & pull activities
- Core strength & trunk control

Technical & Tactical:

Aim to include:

- Specific skill development
- Technique work & practice
- Tactics and race or event strategies
- Running at different paces

Lifestyle & Support:

Aim to improve or use:

- Nutrition
- Rest
- Hydration

- Appropriate clothing & equipment
- Awareness of Clean Athletics

Psycho-Behavioural:

Aim to improve:

- Mental strength and resilience
- Dealing with success and failure or winning and not winning
- Internal pressures and external expectations
- Focus of attention and mental rehearsals.

PERFORMANCE FACTORS - TIME TO REFLECT

Why not take a little time to reflect and ask yourself the following questions? Think about a complete beginner, as well is an experienced runner.

- How well could you address each of the performance factors?

- Are there any areas that you're not confident with?

If you feel you need to learn more, why don't you try working with more experienced leaders or visiting existing groups and clubs.

You may also find more development opportunities and information at the websites below:

https://runtogether.co.uk

https://jogscotland.org.uk

https://irun.wales

https://www.runbritain.com

THE FIVE FITNESS COMPONENTS

Let's look at the five basic fitness components. Have a go at selecting what you think are the five basic fitness components:

- Endurance
- Weight
- Speed
- Balance
- Strength
- Height
- Flexibility
- Coordination
- Agility

Here are the correct answers:

- Endurance
- Speed
- Strength
- Flexibility
- Coordination

Each exercise during training will usually develop a particular fitness component. For example, if the distance or duration of an exercise is increased, then this will develop endurance. During training sessions, you should look to incorporate all the components and incorporate as many into and across your coaching sessions as is appropriate for the age and stage of development of the runner.

FITNESS COMPONENTS IMPROVEMENTS

Thinking of the five fitness components displayed:

Endurance, Flexibility, Speed, Coordination, Strength

- Where can the biggest improvements be made initially?
- What components could runners work on alone?
- When would flexibility and coordination be the focus?

Write your answers down.

Now compare your ideas to our suggestions. Please note that this list is not exhaustive:

- The biggest improvements can be made to all aspects, especially in beginner runners. However, an example, of where initial/quick improvements can be made involve endurance improvements. These can be achieved by increasing the distance or duration of an exercise.
- Runners can work on endurance on their own by increasing the distance or duration of an exercise. They can work on strength and coordination by participating in other exercise such as circuits, Pilates, yoga, weight-training etc.
- Flexibility and coordination can be the focus during warm up and cool down parts of the sessions.

TRAINING PRINCIPLES

Have you ever wondered why training works? Training works because the body can adapt. Adaptation happens after a session and during the recovery. Recovery can be active or passive.

Let's take a look at the training principles in a bit more detail:

- **Overload** – a runner has worked the body hard during a session. This may result in the runner feeling tired with potentially aching muscles. A runner can adapt to loads that are higher than the demands of normal daily activity. By varying the type, volume and intensity of training the runner's body has an opportunity to recover and over-compensate. Loading must increase incrementally to create improvement and avoid injury.

- **Recovery and Adaptation** - after a day or two of appropriate rest, the runner's body starts to repair and grow stronger. Their body prepares itself for the next training session. Adaptations to the demands of training take place gradually, over long periods of time. Rushing this process may put the runner at risk of injury, illness, or overtraining. Allowing the runner to recover and adapt appropriately enables the body to 'super compensate'.

- **Specificity** - training should be relevant and appropriate to the fitness component / event group in order to get the desired effect. For example, this means that you should look to get runners to undertake sprint training, in order for them to develop speed. In addition, a well-rounded training programme, containing all elements of the fitness components and energy system development is recommended, ensuring the activities are appropriate to the stage of development of the runner.

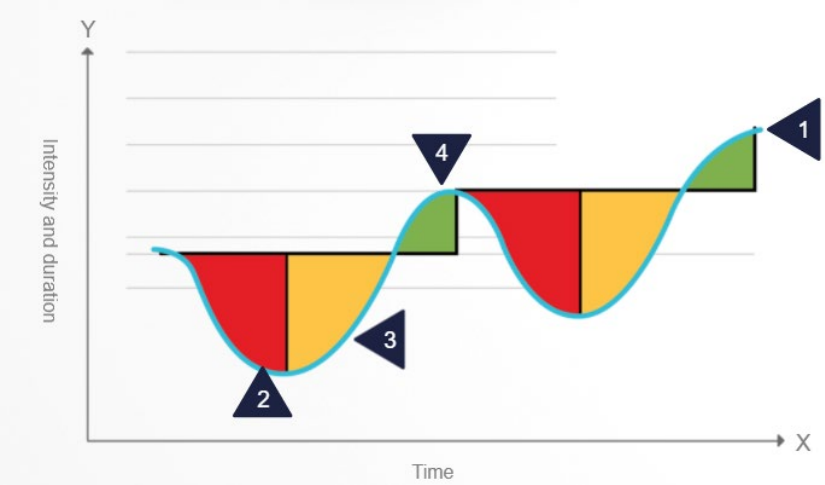
Another important aspect of the Training Principles are the FIT Factors. Training can be varied depending on each or all of these factors (please note, only one of these should be varied at a time):

- **Frequency** - how often the runner trains per week
- **Intensity** - how hard the runner works
- **Time/duration** – how long the runs/sessions are.

The training load is defined by these. Recovery can be active or passive and you have to identify the most appropriate FIT factor for each runner. Take a little time to think about this in relation to a beginner and an experienced runner.

TRAINING PRINCIPLES GRAPH

Why don't we take a look at a graph to help you understand the training principles.



- The blue line represents the performance levels and shows that this is improving over time.
- The red element represents the overload and specific training stimulus.
- The yellow element represents recovery and adaption time.
- The green element shows the super-compensation period and this is the post training period during which the runner has higher performance capacity than before.

You'll see that as a result of introducing the second training stimulus at the peak of the performance level during the 'super compensation' phase, the runner's performance level is higher than at the start of the first training session. This can be considered 'progressive overload' and should see improvements in performance (fitness) levels.

Did you know that the greater the training load (harder the session), the greater the amount of recovery is needed to allow the body to recover, so that it's slightly more prepared for the next session than before?

SUPER-COMPENSATION ADDITIONAL INFORMATION

Here's some additional information on super-compensation:

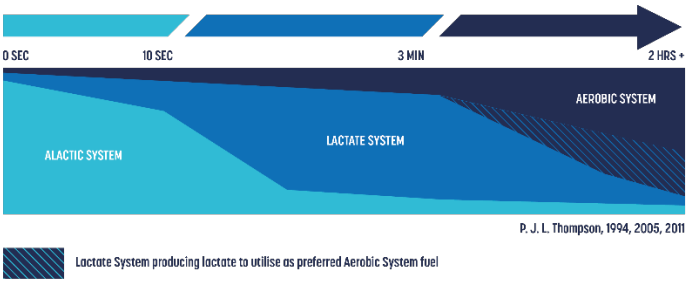
In scientific terms super-compensation is a sports science theory that focuses on the elevated increase of glycogen in the muscles. The theory states that when larger amounts of glycogen are released in the muscles, the body will react with a sudden drop in glycogen levels that is then followed by an increased desire for carbohydrates. Consider the Lifestyle & Support Performance Factor - what food / drinks would you have post training? This chain of events allows the body to compensate for the sudden increase in muscle glycogen and restore balance.

The process of super-compensation is said to follow immediately after the training period, when the body is fatigued from the stress of the training. During this time, natural processes in the body trigger to restore the level of energy and fitness that the individual possessed prior to the beginning of the training, and also prepare the body for another round of training by increasing the general energy level. If the individual does initiate a second round of training while this increased energy level is present, the body gradually adjusts to this new level, and it becomes the basic fitness level for the body in the future.

IDENTIFY THE DIFFERENCES IN THE 3 ENERGY SYSTEMS

Let's take a look at the 3 energy systems. These are:

- aerobic
- alactic
- lactate



Alactic System

- Not dependant on oxygen, plus no lactate or acid is produced

- Immediate highest intensity activity but can only sustain it for a few seconds.
- High quality speed and power work of 3-8 seconds, with 2-3 minutes rest to allow full recovery and replenishment of fuel.

Lactate System

- Oxygen independent and produces lactate and acid.
- It is known as the 'linking' energy system because it provides the bridge between the capabilities of the aerobic and alactic systems.
- Repetition training, fartlek and circuits where higher intensity work is required with partial recovery or short lower intensity periods.
- Improved ability to generate energy from this system and to create and use lactate as an aerobic fuel source.
- For pre-pubescent athletes the lactate system is underdeveloped. Pre-pubescent runners don't fully understand when they are working too hard and we have to protect them from this.

Aerobic System

- Oxygen dependent and uses various fuels, including lactate.
- Prolonged low to moderate intensity work.
- Aerobic endurance training, e.g. steady state running, cycling, swimming for 2-30 minutes or longer.

Let's give you the opportunity to see if you can work out which energy systems you're using during different scenarios

Scenario 1

Which energy system are you predominantly using as you sit there now?

Scenario 2

What about if I asked you to run or move a body part very fast for 60 secs?

Scenario 3

Which energy system would you be using if you had to sprint to stop the saucepan boiling over?

Scenario 4

Which energy system would you be using if you were sprinting the last 5 metres of a 5K race?

Here are the correct answers:

Whilst sitting there you are predominantly using your aerobic energy system.

During the first few seconds of intense activity, you would have predominantly been using the **alactic energy system**. As the intense activity continued, the alactic energy system would have become depleted and your body would have been forced to produce energy using the **lactate energy system**.

To sprint to the saucepan, you'd be using the alactic energy system. This is the 'start up' energy that we can use immediately, until its stores are depleted, after approximately 8-10 seconds of energy.

This utilises the alactic energy that is stored up, ready to be used for a short burst of speed i.e. the finish. It also utilises the aerobic system as this will have been the predominant system for most of the 5K race.

All three energy systems are operating all the time. For pre-pubescent athletes the Lactate system is under-developed and the aerobic system tends to dominate. As the Lactate system is underdeveloped, pre-pubescent runners don't fully understand when they are working too hard and we have to protect them from this.

It is the duration and intensity of activity which determines the predominant energy system.

F.I.T. PRINCIPLES

Let's take a moment to remember the F.I.T. principles.

F FREQUENCY - this is how **often** someone exercises each week.

I INTENSITY - This is how hard the body has to work during exercise.

T TIME - this is how **long** someone exercises in each session.

THE TALK TEST

Have you thought about how you can tell how hard runners are working? Let's take a look at the talk test. By listening to your runners, you can get a good indication of which energy system they are using.

Chat in whole sentences: This is likely to be using the aerobic energy system.

Talk in phrases: This is likely to be using aerobic/lactate energy system during moderate intensity activity.

One word: This is likely to be using lactate/alactic energy systems and during vigorous intensity activity.

Next time you are with runners carefully observe how they talk, in order to get any idea of their level of intensity.

ATHLETE DEVELOPMENT

Let's now look at what the Athlete Development Model looks like whilst applying the training principles and fitness components.

A runner should start with the fundamentals if they:

- are an inexperienced runner
- lack basic movement skills
- this is not exclusively for young runners

Fundamental training focuses on:

- warmups

- flexibility
- coordination
- agility and balance
- basic skill introduction
- fun

Considerations for runners at the Foundation stage:

- a returning or a developing runner
- a runner learning to train
- building on ABC's
- competition is limited

Foundation stage training builds on the Fundamentals and develops:

- drills
- technical skills
- running longer, stronger and faster

A runner will be in the event group if:

- they are physically more prepared for activity
- competition/outcome focus is more significant
- they are looking to refine skills and introducing tactics

Event Group training focuses on:

- performance goals
- physical development
- skills development
- psychological development

PROGRESSIVE OVERLOAD AND VARIETY INTRODUCTION

We looked at the training principles and F.I.T Factors to help get a better understanding of progressive overload and to develop the components of fitness. We also covered energy systems and the importance of variety.

Let's see how what we have covered. How does this relate to delivering a series of linked sessions to ensure progressive overload and variety?

Take a look at a **Couch to 5K** running programme, focusing on week **1, 3** and **5** of this 9 week programme.

Week 1

For your 3 runs in week 1, you will begin with a brisk 5-minute walk, **then alternate 1 minute of running and 1-and-a-half minutes of walking, for a total of 20 minutes.**

Week 3

For your 3 runs in week 3, you will begin with a brisk 5-minute walk, **then 2 repetitions of 1-and-a-half minutes of running, 1-and-a-half minutes of walking, 3 minutes of running and 3 minutes of walking.**

Week 5

There are 3 different runs this week:

Run 1: a brisk 5-minute walk**, then 5 minutes of running, 3 minutes of walking, 5 minutes of running, 3 minutes of walking and 5 minutes of running.**

Run 2: a brisk 5-minute walk, then 8 minutes of running, 5 minutes of walking and 8 minutes of running.

Run 3: a brisk 5-minute walk, **then 20 minutes of running, with no walking**.

You now have a better understanding of how the **Couch to 5K** uses progressive overload and variety. Here's some additional information:

During **week 3**, there's an increase in length of time running by 30 seconds for 2 of the repetitions, but with no change in length of time for the recovery walk. The other repetitions have an increase in time spent running of 2 minutes from week one, but this is accompanied by an increase in the recovery walk time.

During **week 5**, **run 1** is a non-continuous running session as it has recovery walks planned in. **Run 3** is a continuous running session as there are no recovery walks planned in.

More information on Couch to 5K can be found by clicking here.

SESSION CARDS

We would now like you to go to the resources and open the session cards and take some time to read them.

Taking into consideration the principles of **variety** and **progressive overload**, see if you can identify which cards support the delivery of a **continuous running** fitness activity and which support **non-continuous running** fitness activities. For each session type, click in the **continuous** or **non-continuous** check box.

- Fartlek
- Indian File
- Fixed point repetitions
- Raid the goal
- Out and back runs
- Hills running (pyramids)
- Meet and repeat
- The loop back
- Pairs running or relays

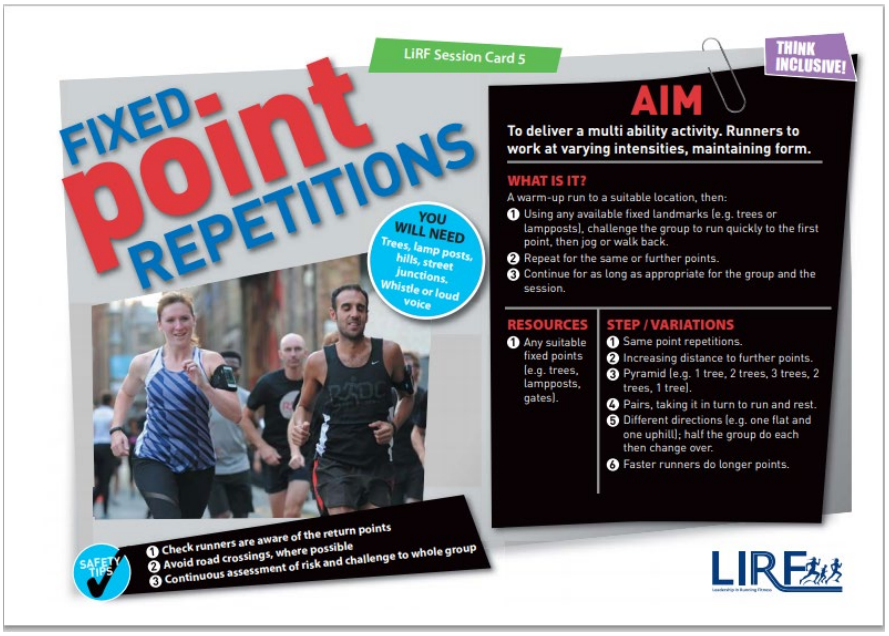
- Figure of eight
- Figure of eight relay
- Timed loops

Here are the correct answers.

- Fartlek is **continuous**
- Indian File is **continuous**
- Fixed point repetitions is **non-continuous**
- Raid the goal is **non-continuous**
- Out and back runs is **continuous**
- Hills running (pyramids) is **non-continuous**
- Meet and repeat is **continuous**
- The loop back is **continuous**
- Pairs running or relays is **non-continuous**
- Figure of eight is **continuous**
- Figure of eight relay is **non-continuous**
- Timed loops is **continuous**

FIXED POINT REPETITION CARD 1

We'd now like you to take a look at the **Fixed point repetition** card.



Can you see where **progressive overload** is suggested as a **variation**?

Step variation 2 offers progressive overload as a variation by **increasing** the distance to further points.

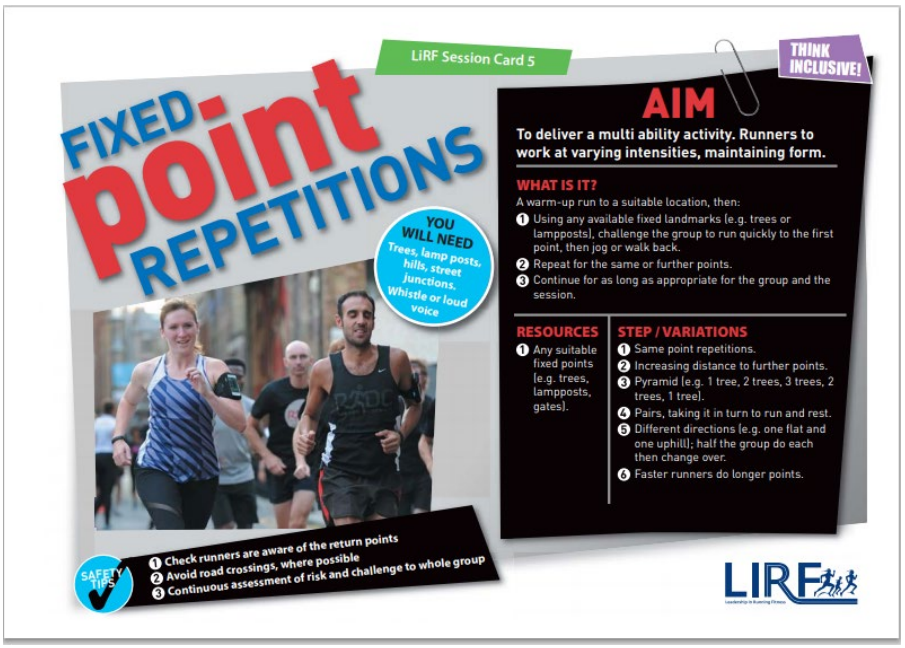
Step 3 variation offers progressive overload as a variation by **increasing** the number of trees in the pyramid.

Step 5 variation is offering progressive overload as a variation by **increasing** the challenge through the introduction of a hill.

You'll see that this also delivers a multi-ability session, as it enables runners to return to the first point at their own pace. For faster runners, you can arrange longer running points. Throughout all of your sessions you should be continuously assessing risks and challenge to all individuals.

FIXED POINT REPETITION CARD 2

Let's take another look at the same **Fixed point repetition** card.



Can you see where **variety** and **change in intensity** is suggested as a **variation?**

Step variation 4 allows pairs taking turns to run and rest. Resting before running the same distance again is likely to encourage a **higher intensity** than repeating the repetitions with no rest.

Step variation 5 also challenges runners over different gradients, one on the flat and one uphill. Hills of the same distance as the flat run**, will increase the intensity** of the activity.

PLANNING THE FIRST 6 WEEKS

Have you thought about what you should consider when planning your first 6 weeks with your group? Here are some questions to consider.

- Who are you likely to be leading?
- How many sessions will you be leading per week?
- Will you have the group for a period of time?

- Do they have individual or a group goal?
- What are your own limitations as a leader?

Take some time to gather answers to these questions to help you with your planning.

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- What are your own limitations as a leader?

Take some time to gather answers to these questions to help you with your planning.

Here are some final tips to take away:

- Remember to repeat good practice and go about developing good safety.

- Why not take some time to look through the 12 session cards and have a think about planning out your first 6 weeks of training. Ensure that sessions provide variety and consider the various needs of the group. You also need to plan how runners will progress during the 6 week block.

- Depending on the ability levels within your group, you may want to consider using the different RunTogether programmes (get me started; keep me going; challenge me). These also refer to the session cards we have introduced.

- If you require further assistance with planning sessions then a qualified coach or RunLeader mentor may be able to assist. You may also see the HCAFS programmes and the web addresses can be found below:

  Runtogether.co.uk
  Jogscotland.org.uk
  Irun.wales
  Runbritain.com

CONGRATULATIONS

Congratulations! You have now finished this British Athletics - Leadership in Running Fitness Course - Factors that affect performance and energy systems and their application.

You should now be able to:

- Describe the factors that influence performance
- Identify the components of fitness
- Identify how training works and differences in energy systems
- Identify how to progress a groups running fitness

- Describe how to adapt a session for a mixed ability group to vary intensity or reflect individual needs

Thank you.